WORKED SOLUTION

STEP II 2024, Question 5

Pure • Ranges and Diophantine Equations

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Read the question before you go on. It is in the STEP II paper in OCR's download of the 2024 STEP papers (a zip file):

www.ocr.org.uk/Images/725753-2024-past-papers.zip

The key idea

The range of a polynomial on $\mathbb{Z}$ doesn't change if we shift the variable by an integer, so we complete the square (or the cube) to put each function in its simplest shifted form. Asking whether two ranges meet then becomes asking whether an equation in two integer unknowns has a solution, and the trick is to rearrange that equation into the form *product of two integers = a small fixed number*. Factorising a small number has only a handful of possibilities, so we can check each one.

Solution

Part (i)

Completing the square,

$$\mathrm{f}_1(n) = (n+3)^2 + 2 = \mathrm{F}_1(n+3).$$

As n runs through $\mathbb{Z}$, so does $m = n+3$ (every integer m arises, from $n = m-3$). So the range of f_1 is $\{m^2+2 : m \in \mathbb{Z}\}$, which is exactly the range of F_1. That is what we needed.

Part (ii)

Similarly $\mathrm{g}_1(n) = (n-1)^2 + 4$, so by the same argument the range of g_1 is $\{k^2+4 : k \in \mathbb{Z}\}$. By part (i), a common value would need integers m, k with

$$m^2 + 2 = k^2 + 4 \quad \Longleftrightarrow \quad (m-k)(m+k) = 2.$$

Now $m-k$ and $m+k$ differ by $2k$, which is even, so they have the same parity. If both are odd, their product is odd, so it can't be 2. If both are even, their product is a multiple of 4, so again it can't be 2. Hence no such m, k exist, and the two ranges have empty intersection.

Part (iii)

Completing the square,

$$f_2(n) = (n-1)^2 - 7, \qquad g_2(n) = (n-2)^2 - 2,$$

so the ranges are $\{m^2 - 7\}$ and $\{k^2 - 2\}$ with m, k running over $\mathbb{Z}$. A common value needs

$$m^2 - 7 = k^2 - 2 \quad \iff \quad (m-k)(m+k) = 5.$$

Since 5 is prime, the pair $(m-k, m+k)$ is one of $(1,5)$, $(5,1)$, $(-1,-5)$, $(-5,-1)$. Adding the two factors gives $2m = \pm 6$, and subtracting gives $2k = \pm 4$, so $m^2 = 9$ and $k^2 = 4$ in every case. The common value is

$$m^2 - 7 = 9 - 7 = 2 = 4 - 2 = k^2 - 2.$$

So the only integer in both ranges is $\boxed{2}$. As a check, $f_2(4) = f_2(-2) = 2$ and $g_2(4) = g_2(0) = 2$.

Part (iv)

Completing the square in p,

$$p^2 + pq + q^2 = \left(p + \tfrac{1}{2}q\right)^2 + \tfrac{3}{4}q^2 \geq 0,$$

since both terms are squares times positive constants. (Equality holds only when $q = 0$ and $p = 0$.)

For the cubics, we compare f_3 with $(n-1)^3 = n^3 - 3n^2 + 3n - 1$:

$$f_3(n) = (n-1)^3 + 4n + 1 = (n-1)^3 + 4(n-1) + 5.$$

So, writing $m = n - 1$ (which again runs over all of $\mathbb{Z}$), the range of f_3 is $\{m^3 + 4m + 5\}$, and the range of g_3 is $\{k^3 + 4k - 6\}$. A common value needs integers m, k with

$$\begin{aligned} & m^3 + 4m + 5 = k^3 + 4k - 6 \\ \iff \quad & (k^3 - m^3) + 4(k-m) = 11 \\ \iff \quad & (k-m)\left(k^2 + km + m^2 + 4\right) = 11. \end{aligned}$$

Ask yourself How does the inequality at the start of this part help with this equation?

By the inequality with $p = k$, $q = m$, we have $k^2 + km + m^2 + 4 \geq 4$. In particular the second factor is a positive integer, so $k - m$ must be positive too. Since 11 is prime, there are only two options:

- $k - m = 11$ and $k^2 + km + m^2 + 4 = 1$. This is impossible, because the second factor is at least 4.

- $k - m = 1$ and $k^2 + km + m^2 + 4 = 11$.

In the second case, put $k = m + 1$:

$$\begin{aligned}(m+1)^2 + m(m+1) + m^2 &= 7\\ 3m^2 + 3m + 1 &= 7\\ m^2 + m - 2 = (m+2)(m-1) &= 0,\end{aligned}$$

so $m = 1$ (with $k = 2$) or $m = -2$ (with $k = -1$). The corresponding values are

$$m = 1 :\ 1 + 4 + 5 = 10, \qquad m = -2 :\ -8 - 8 + 5 = -11.$$

So the integers in both ranges are $\boxed{10 \text{ and } -11}$.

Returning to the original variables: $\mathrm{f}_3(2) = 8 - 12 + 14 = 10$ and $\mathrm{g}_3(2) = 8 + 8 - 6 = 10$; also $\mathrm{f}_3(-1) = -1 - 3 - 7 = -11$ and $\mathrm{g}_3(-1) = -1 - 4 - 6 = -11$. Interestingly, both common values come from the same input n.

Checks

- Substitute back, as we did above: f_2 and g_2 both take the value 2 at $n = 4$, and f_3, g_3 agree at $n = 2$ (value 10) and $n = -1$ (value -11).
- Listing small values is a quick sanity check for part (ii): f_1 gives $2, 3, 6, 11, 18, 27, \ldots$ and g_1 gives $4, 5, 8, 13, 20, 29, \ldots$. Neither list contains a number from the other.
- A computer search over all $|n| \leq 3000$ confirmed that the ranges in (i) agree, the ranges in (ii) are disjoint, and the only common values are 2 in (iii) and 10, -11 in (iv).

Common slips

- Finding a common value by solving $\mathrm{f}(n) = \mathrm{g}(n)$ for the *same* n. The two inputs can be different, so we need two independent unknowns. In (iii), $\mathrm{f}_2(n) = \mathrm{g}_2(n)$ happens to give the right answer, but the method is wrong.
- In (ii), arguing only that $(m - k)(m + k) = 2$ forces $\{m - k, m + k\} = \{1, 2\}$ and forgetting the negative factor pairs. The parity argument deals with every case at once.
- In (iv), forgetting to rule out the factorisation 11×1. It's the inequality that does this.

Beyond STEP

Equations like $m^2 - k^2 = 5$ and $k^3 + 4k = m^3 + 4m + 11$, where only integer solutions count, are *Diophantine equations*. Factorising one side and using the divisors of the other

is one of the most powerful elementary tools for solving them. The parity trick in (ii) generalises: an integer is a difference of two squares exactly when it is odd or a multiple of 4. For more, Davenport's *The Higher Arithmetic* is an excellent first-year introduction to number theory.

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